

Column Stability & Equivalent Inertia Summary Report

1. Overview of the Eigenvalue Formulation

In structural stability analysis, the exact differential equation for a column under axial load P with general end restraints yields a boundary value problem. For a uniform column, the displacement function is expressed as:

$$v(z) = A + Bz + C \sin(kz) + D \cos(kz) \quad \text{where } k = \sqrt{\frac{P}{EI}}$$

Applying boundary conditions with rotational springs (α_T, α_B) and translational springs (β_T, β_B) leads to the 4×4 transcendental system determinant:

$$\begin{vmatrix} T_T & (kL)^2 & 0 & T_T \\ 0 & R_T & R_T kL & (kL)^2 \\ T_B & [T_B - (kL)^2] & T_B \sin(kL) & T_B \cos(kL) \\ 0 & R_B & [R_B kL \cos(kL) - (kL)^2 \sin(kL)] & [-R_B kL \sin(kL) - (kL)^2 \cos(kL)] \end{vmatrix} = 0$$

Where the dimensionless parameters are defined as:

$$R_T = \frac{\alpha_T L}{EI}, \quad R_B = \frac{\alpha_B L}{EI}, \quad T_T = \frac{\beta_T L^3}{EI}, \quad T_B = \frac{\beta_B L^3}{EI}$$

2. Integration of Equivalent Inertia (I_{eff}) for Variable Section Columns

For columns with non-uniform or variable cross-sections (e.g., tapered, haunched, or stepped columns), $I(z)$ varies along the length L . To avoid complex numerical discretization or variable-coefficient differential equations, an **Equivalent Inertia** (I_{eff}) approach is integrated directly into the determinant matrix model.

Step 1: Compute Equivalent Inertia (I_{eff})

Determine I_{eff} based on the cross-section variation profile:

- **Linearly Tapered I-Sections (AISC Approach):**

$$I_{\text{eff}} = C_x \cdot I_{\text{min}}$$

where $C_x = 1.0 + 0.08\gamma + 0.8\gamma^2$ and $\gamma = \frac{d_{\text{max}}}{d_{\text{min}}} - 1$.

- **Stepped Columns:**

$$I_{\text{eff}} \approx I_1 \left(\frac{a}{L} \right) + I_2 \left(\frac{b}{L} \right)$$

Step 2: Update Dimensionless Ratios

Substitute I_{eff} into all boundary stiffness ratios:

$$R_T = \frac{\alpha_T L}{EI_{\text{eff}}}, \quad R_B = \frac{\alpha_B L}{EI_{\text{eff}}}$$
$$T_T = \frac{\beta_T L^3}{EI_{\text{eff}}}, \quad T_B = \frac{\beta_B L^3}{EI_{\text{eff}}}$$

Step 3: Solve for Eigenvalue kL

Find the non-trivial root kL that satisfies $\det[M(kL)] = 0$. The physical parameter k is defined in terms of I_{eff} :

$$k = \sqrt{\frac{P}{EI_{\text{eff}}}}$$

Step 4: Calculate Critical Buckling Load (P_{cr}) & Effective Length (K)

Once kL is obtained, calculate the critical buckling load:

$$P_{\text{cr}} = \frac{(kL)^2 EI_{\text{eff}}}{L^2}$$

The effective length factor K relates directly to the eigenvalue kL by:

$$K = \frac{\pi}{kL}$$

Structural Stability & Elastic Buckling Analysis Reference